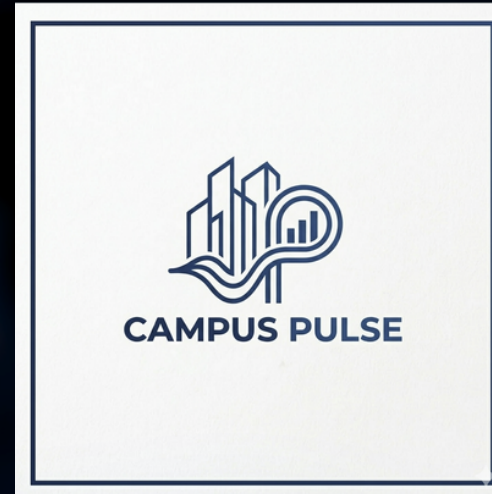




CAMPUS PULSE

TEAM 4

C. JUNWHAN, K. HENRY, L. SUYEON, N. BYEONGMIN,

CURRENT ISSUE



- **No visibility of real-time building occupancy**
 - Floor-level crowding information is unavailable
 - Libraries, dining halls become overcrowded
 - Varying waiting time throughout the day





CONSTRUCTION

RECOGNITION OF ISSUE

- Primary extraction of information



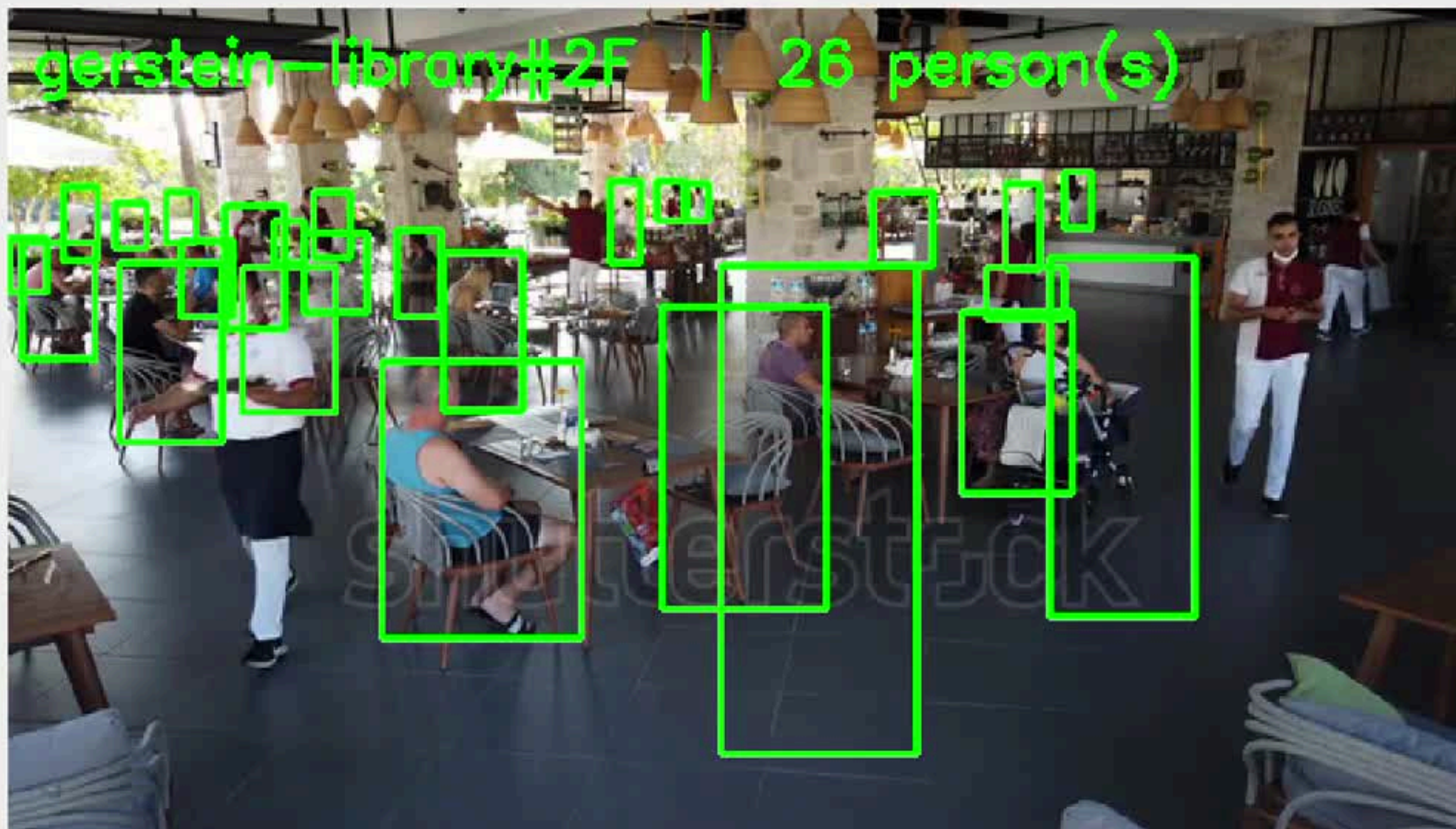
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AMAZON REKOGNITION

- Occupation detection using camera based and tools by Amazon Rekognition
- Converts video information into numerical headcount data





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- Stores and updates data



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- Views data from DynamoDB
- Organize data
- Generates API response



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API GATEWAY & REACT

- Summon API
- Visualise data
- Real-time update

CAMPUS PULSE

- Real-time building density
 - Displayed in percentages and colour
- Availability of low-density facilities
- Crowd trend over time



CAMPUS PULSE

CP UofT Smart Campus
Campus Pulse

Live Campus Occupancy Dashboard

Building-level crowd status and floor-by-floor density. Designed for quick scanning and one-click building details.

Mock ModeAPI ModeRefresh

Best location right now

Sidney Smith 20% occupied

⚠ Active building alerts

Gerstein: Temporary elevator disruption reported. Use alternate route.

Supported Buildings

Click any building to open its detail page.

Building Directory

Choose a building to see its information page, floor details, and daily trend.

Most crowded

LowMidHigh

Robarts
Robarts Commons
Sun 10AM–12AM • Mon–Thu 24 Hours • Fri 12AM–11PM • Sat 9AM–10PM

81%
Updated Just now
High

Bahen
Bahen Centre for Information Technology
General access often listed around 8AM–6PM; after-hours access may require authorization

52%
Updated Just now
Mid

Robarts Commons

Live building info, floor density, and daily trend.

← Back

Updated Just now

Robarts
Study spaces filling quickly.

Study SpaceQuiet ZonesGroup Rooms

CURRENT OCCUPANCY

64%

MidStable

⌚ Operating hours

Sun 10AM–12AM • Mon–Thu 24 Hours • Fri 12AM–11PM • Sat 9AM–10PM
Robarts Common overnight access runs Sunday to Thursday during fall and winter terms.

Floor Details

Per-floor density snapshot for the selected building.

Cards

Table

Floor

1F

Mid

57%

Updated Just now

Density

Floor

2F

High

96%

Updated Just now

Density

Floor

3F

High

84%

Updated Just now

Density

Floor

4F

Low

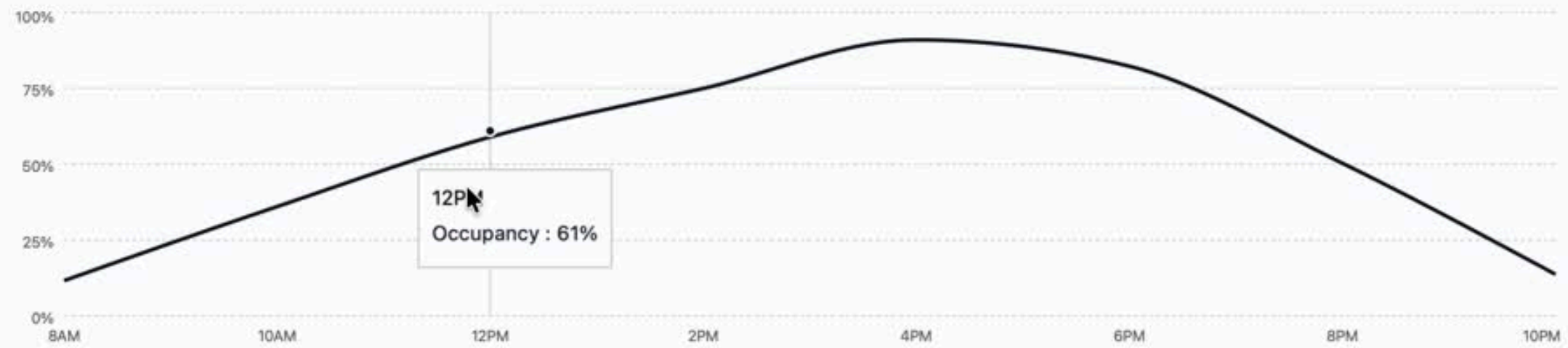
35%

Updated Just now

Density

Today's Occupancy Trend

See when this building is busier or quieter during the day.

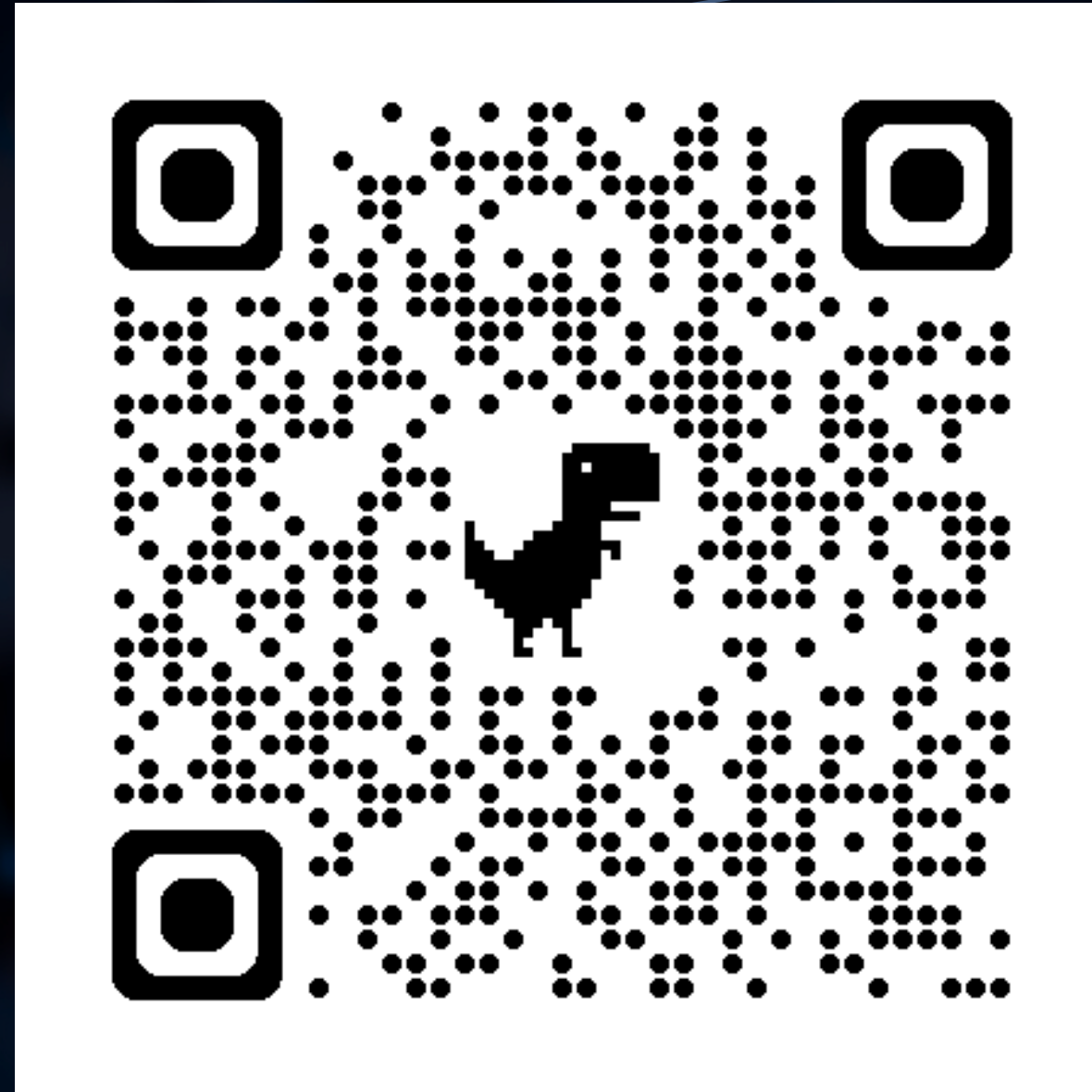




FUTURE STEPS

- Updates & status on room bookings in libraries
- GPS based map to display routes, distances to each facility from the user
- AI for trend analysis & prediction
- Alert system for emergencies → link to ambulances/Fire department for population density
- Expand to other facilities such as dining halls, rest spaces

CAMPUS PULSE



QR Code for the Campus Pulse Demo Website

Hosted via **AWS API Gateway**



THANK YOU